

Observability Limits of Learned Satellite Cyber-Recovery Decisions Under Compromised Evidence

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Abstract

Cyber-recovery decisions can remain unsafe even when the evidence presented to a decision policy is authenticated, current, and internally consistent, because a compromised evidence producer can supply a plausible but false state. This study evaluates whether a learned satellite cyber-recovery selector can overcome that observability boundary when it receives the same policy-visible inputs as a deterministic comparator, and what changes when a corroborating observable is added. A deterministic finite assurance experiment exhaustively evaluates 1,033 modeled observations: 512 observations over all 256 eight-feature visible states for a fixed comparator and a visible-only learned selector, 512 observations over all 512 nine-feature states for a corroboration-aware learner, and nine prespecified hidden-truth collision observations. The visible-only learner reproduces the complete visible decision boundary with zero errors, yet it proceeds unsafely when hidden authorization changes while its inputs remain unchanged. The corroboration-aware learner resolves the prespecified independent-disagreement collision, but it again proceeds unsafely when corroboration shares the false state; across its complete nine-feature lattice it has two objective-decision errors, one unsafe proceed and one false-conservative hold. The results show that model fit cannot substitute for decision-relevant observability and that corroboration shifts, rather than removes, the trust boundary of learned recovery logic.

Keywords: Satellite cybersecurity; cyber recovery; machine learning assurance; observability; evidence integrity; spacecraft autonomy

1 Introduction

Cyber recovery is a different decision problem from cyber detection. Detection can raise an alarm or identify an anomalous condition; recovery logic must decide what action remains justified after that alarm using the state visible to the decision process. Space-domain concepts such as cyber-safe mode similarly connect cyber-threat response to controlled transition toward an integrity-protected recovery baseline [1]. The distinction becomes important when evidence used to authorize recovery is itself exposed to compromise. A recovery selector can receive a record that remains authenticated and current enough for policy qualification while still being false relative to a state that the selector cannot observe.

A prior frozen study in this research program evaluated that condition directly. Under its V5 treatment, a controlled policy-visible producer could supply a validly signed, current value that remained acceptable to the recovery gate while disagreeing with research-only adjudication truth. The resulting boundary was not a cryptographic verification failure: the record could satisfy the implemented qualification checks while hidden authorization was false. The Study-2 source-evidence package is publicly archived on Zenodo [2]. The present article does not reuse Study-2 observations or statistical results; it uses the signed-but-false condition and deterministic selector semantics only as antecedent specification for a separately designed and frozen machine-learning assurance experiment.

The question is whether a learned selector changes that boundary when it receives the same information. Exact fit to a visible decision rule does not itself provide hidden information absent from the model input. Conversely, adding another observable changes the information available to the decision process, but any benefit depends on the failure relationship between that observable and the compromised evidence path. This distinction is relevant to aerospace assurance, where confidence in a learning-enabled component depends on evidence concerning functional intent, model and data properties, implementation, integration, coverage, traceability, robustness, and verification and validation in addition to model performance [3].

This article reports experiment S7-LSO-001, a deterministic finite study comparing a fixed visible-only recovery rule, a learned selector trained on the same eight policy-visible inputs, and a second learned selector that receives one

additional corroboration bit. The complete modeled population contains 1,033 exact observations and is checked by a separately implemented repository audit that reports zero mismatches. The primary research question is: How does policy-visible evidence constrain the adjudicated correctness of a learned satellite cyber-recovery selector under signed-but-false evidence, and how does adding a corroborating observable change that boundary when corroboration is independent versus correlated with the compromised evidence path?

The contribution is intentionally bounded. The study exhausts the frozen visible and corroboration state spaces; tests whether exact learning of a visible decision boundary implies correctness when relevant truth is hidden; isolates the effect of an added corroborating observable under independent and correlated failure; and preserves hash-frozen execution plus a separately implemented audit record. It does not claim a new general theory of partial observability, operational spacecraft autonomy, certification sufficiency, or global superiority of learned over deterministic recovery logic.

2 Related work and assurance boundary

Satellite fault detection, isolation, and recovery (FDIR) is a mature spacecraft-autonomy problem, but recent work approaches it through mechanisms that differ materially from the trust question studied here. Wanninger [4] presents a CLIPS-based expert-rule FDIR component integrated with a Packet Utilization Standard-oriented onboard software architecture, emphasizing flexible rule representation and in-flight modification of failure-handling logic. That work demonstrates structured recovery implementation; it does not model adversarially false trusted evidence or ask whether the policy-visible state is sufficient to distinguish safe from unsafe recovery authorization.

AI-enabled space fault management has also advanced beyond anomaly detection alone. Tappe et al. [5] integrate supervised anomaly detection, diagnosis, and reconfiguration support for the COLUMBUS module of the International Space Station, combining data-driven and knowledge-based components in a practical fault-management toolchain. That work is the closest CEAS-native neighbor to the present study because it connects AI to diagnosis and reconfiguration. Its research question, however, concerns detecting and managing physical-system faults rather than assuring a cyber-recovery authorization decision when the evidence path itself may be compromised.

A second adjacent literature focuses on telemetry data and anomaly-detection performance. The OPS-SAT benchmark provides real spacecraft telemetry fragments and labels for comparing anomaly-detection methods [6]; transformer-based work evaluates competing architectures on satellite telemetry [7]; and validation-gated continual learning addresses telemetry drift and data scarcity [8]. Those studies improve detection, prediction, or adaptation. The present study begins downstream of detection: its `security_signal` variable is already part of the policy-visible state. It asks whether the information presented to the recovery selector is sufficient for a correct authorization decision.

The novelty is therefore application-specific rather than a claim that partial observability or corroboration is new. A visible-only learned selector is trained against the same policy-visible decision semantics as a deterministic comparator and evaluated over the entire eight-feature state space. A second learned selector receives one corroboration observable and is evaluated over the complete nine-feature state space. A hidden-truth collision block then holds the original eight visible inputs constant while changing research-only authorization truth and corroboration. No observation from Study 2 or any other submitted study enters the 1,033-observation population.

3 Study design

3.1 Visible state and adjudication

The study uses eight binary policy-visible inputs: `signature_valid`, `source_trusted`, `fresh`, `epoch_valid`, `noncontradictory`, `minimum_evidence_complete`, `security_signal`, and `authorization_available`. The first six variables express evidence-quality conditions. `security_signal` identifies the modeled security-response condition, and `authorization_available` is the authorization state visible to the policy. The corroboration-aware learner receives one additional binary feature, `independent_corroboration`.

A separate `hidden_authorization` value is used only by the research adjudicator when computing whether it is objectively safe to proceed. It is never supplied to the deterministic comparator or either learned selector. When no hidden-truth collision is imposed, hidden authorization equals visible authorization. In the collision block, the two may differ. The adjudicated objective is safe-to-proceed only when all six evidence-quality conditions hold and either no

security signal is present or adjudicated authorization is true. This construction prevents the learner from receiving the research-only answer through a privileged feature.

3.2 Decision policies and deterministic learning procedure

Three policies are evaluated. D0_S1_VISIBLE_ONLY is the fixed visible-only comparator.

L0_ERM_VISIBLE_ONLY is learned from the same eight visible inputs.

L1_ERM_WITH_INDEPENDENT_CORROBORATION adds the corroboration bit.

Both learned policies use deterministic empirical-risk minimization over a bounded integer linear-threshold hypothesis class. The visible-only model has the form $q \cdot \text{sum}(\text{quality6}) + w_a \cdot \text{authorization} + w_s \cdot \text{security} \geq \text{threshold}$, and the corroboration-aware model adds $w_c \cdot \text{independent_corroboration}$. The frozen protocol uses 10 visible-only training prototypes and 13 corroboration training prototypes. Candidate models are selected by minimum training error, then minimum absolute weight sum, followed by deterministic lexicographic tie-breaking over threshold and weights. No external machine-learning library, stochastic optimizer, reinforcement-learning environment, neural network, or adversarial perturbation optimizer is used. The simplicity is intentional: the study isolates observability and evidence sufficiency rather than model capacity.

The accepted execution produced the frozen models summarized in Table 1. L0_ERM_VISIBLE_ONLY has quality weight 2, authorization weight 1, security weight -1, threshold 12, and zero training errors.

L1_ERM_WITH_INDEPENDENT_CORROBORATION has quality weight 2, authorization weight 0, corroboration weight 1, security weight -1, threshold 12, and zero training errors.

Table 1 Frozen policy and information boundary

Policy	Visible inputs	Learned?	Frozen parameters	Hidden authorization visible?
D0	Eight base features	No	Fixed visible rule	No
L0	Eight base features	Yes	$q=2; w_a=1; w_s=-1; \text{threshold}=12$	No
L1	Base features + corroboration	Yes	$q=2; w_a=0; w_c=1; w_s=-1; \text{threshold}=12$	No

3.3 Finite evaluation blocks and endpoints

Block A exhausts all $2^8 = 256$ combinations of the eight policy-visible features for both D0_S1_VISIBLE_ONLY and L0_ERM_VISIBLE_ONLY. Hidden authorization equals visible authorization in this block. The block therefore contains 512 observations and tests whether the learned selector reproduces the complete visible decision boundary rather than only its training prototypes.

Block B exhausts all $2^9 = 512$ combinations of the eight base features and the corroboration bit for L1_ERM_WITH_INDEPENDENT_CORROBORATION. Hidden authorization again equals visible authorization. This block exposes the complete decision surface created by adding corroboration to the policy-visible state.

Block C holds the eight-feature base vector at a fully qualifying visible state while research-only authorization truth and corroboration vary across three prespecified scenarios: SAFE_CORROBORATED, V5_INDEPENDENT_DISAGREEMENT, and V5_CORRELATED_FALSE_CORROBORATION. Each scenario is evaluated by all three policies, yielding nine observations. The complete finite population is therefore $512 + 512 + 9 = 1,033$ exact observations.

Primary endpoints are objective_decision_error, unsafe_proceed, false_conservative_hold, hidden-truth collision indistinguishability, and resolution of the independent-disagreement case by the corroboration-aware learner. Counts over these states are finite-population results, not estimates of operational spacecraft error probabilities.

3.4 Freeze, provenance, and implementation-independent audit

The accepted execution is bound to GitHub Actions workflow run 33689625480 and commit

f1530b0b2e81a5916adaf7ce808075156424dfb5. The frozen evidence artifact is identified by SHA-256

26b522d2692516aa1b4ae62d032a9a909dbaccd7ef1a74104080c39bdf3a091b. The four core deterministic outputs are

separately hash-bound in the repository results freeze: observations.csv, policy_summary.csv, TRAINED_MODELS.json, and REPORT.json.

A separately implemented audit reconstructs the objective function, both learned models, and all 1,033 policy decisions from the frozen evidence without importing or invoking the primary analyzer, and it reports zero mismatches. This is implementation independence within the repository, not an independent human replication. Manuscript preparation does not rerun the experiment, retrain either learner, or substitute any frozen result.

3.5 AI-assisted manuscript preparation and verification

OpenAI ChatGPT, accessed through the ChatGPT web interface, was used during Study 7 development and manuscript preparation as an interactive research, coding, verification, and writing assistant. Before the results freeze, it assisted with protocol refinement, code and test drafting, repository and audit documentation, and verification-oriented review. After the results freeze, it assisted with manuscript structuring and drafting, literature-source discovery and verification, compliance checking, and language refinement. The author made and approved all research decisions, controlled the repository and experimental execution, reviewed and accepted code and test changes, verified the frozen outputs and numerical claims against the repository evidence, and reviewed and edited all AI-assisted text. ChatGPT was not an author and did not alter the accepted 1,033-observation population or frozen outputs after acceptance. All scientific claims and the submitted manuscript remain the sole responsibility of the author.

4 Results

4.1 Visible-state equivalence and corroboration lattice

In Block A, D0 and L0 each contributed 256 observations. Both produced zero objective-decision errors, zero unsafe proceeds, zero false-conservative holds, and exactly three proceed decisions. L0 therefore reproduced the complete visible decision boundary rather than only its 10 training prototypes.

Block B evaluated L1 over all 512 extended visible states. L1 produced six proceed decisions and 506 holds, with exactly two objective-decision errors: one unsafe proceed and one false-conservative hold. Both errors occurred in the fully qualifying evidence-quality state with security_signal = 1 when visible authorization and corroboration disagreed. With visible authorization 0 and corroboration 1, L1 proceeded although adjudicated authorization was false. With visible authorization 1 and corroboration 0, L1 held although adjudicated authorization was true. These are exact state counts, not estimated error rates.

Table 2 Exact finite-population outcomes

Block	Policy	N	Errors	Unsafe proceed	False-conservative hold	Proceed decisions
A	D0	256	0	0	0	3
A	L0	256	0	0	0	3
B	L1	512	2	1	1	6
C	D0	3	2	2	0	3
C	L0	3	2	2	0	3
C	L1	3	1	1	0	2

4.2 Hidden-truth collisions

Block C keeps all eight original policy-visible inputs equal to one. In SAFE_CORROBORATED, hidden authorization and corroboration are also one; the adjudicated objective is proceed, and D0, L0, and L1 all proceed correctly.

In V5_INDEPENDENT_DISAGREEMENT, visible authorization remains one but hidden authorization is zero. D0 and L0 receive exactly the same eight inputs as in the safe case and both proceed unsafely. L1 receives corroboration zero and holds, matching the adjudicated objective.

In V5_CORRELATED_FALSE_CORROBORATION, hidden authorization remains zero while corroboration returns to one. D0 and L0 again proceed unsafely, and L1 also proceeds unsafely. Adding corroboration therefore resolves the

prespecified disagreement case only while the added observable distinguishes the hidden-false condition. When corroboration shares the false state, the unsafe decision returns.

Together, the blocks answer the research question in two parts. Exact learning of the visible decision boundary does not resolve a hidden-truth collision when the model inputs are identical. Adding a corroborating observable changes the information available to the selector and resolves the prespecified independent-disagreement case, but the complete Block-B lattice and the correlated-failure collision show that the added observable introduces its own trust dependency rather than eliminating the evidence boundary.

Table 3 Prespecified hidden-truth collision scenarios

Scenario	Hidden authorization	L1 corroboration	Objective	D0	L0	L1
Safe corroborated	1	1	Proceed	Proceed - safe	Proceed - safe	Proceed - safe
Independent disagreement	0	0	Hold	Proceed - unsafe	Proceed - unsafe	Hold - correct
Correlated false corroboration	0	1	Hold	Proceed - unsafe	Proceed - unsafe	Proceed - unsafe

5 Discussion

The central result concerns information sufficiency, not model performance. L0 achieved zero training error and zero error over all 256 visible states, yet it had no protection in the hidden-truth collision because the relevant distinction was not in its input. In the independent-disagreement case, D0 and L0 saw the same fully qualifying vector as in the safe case; their unsafe decisions therefore reflect an observability boundary rather than a failure to fit the visible rule.

L1 changed the outcome because the input changed. A zero corroboration bit distinguished the hidden-false disagreement state from the safe corroborated state, allowing L1 to hold. This is not evidence that learned logic is intrinsically safer than deterministic recovery logic. It shows instead that decision correctness can depend on whether a decision-relevant observable distinguishes the hidden condition being adjudicated.

The complete nine-feature lattice prevents a one-sided interpretation of corroboration. L1's two errors occurred when corroboration and visible authorization disagreed in an otherwise fully qualifying security state. Positive corroboration with false adjudicated authorization produced an unsafe proceed; absent corroboration with true adjudicated authorization produced a false-conservative hold. Corroboration therefore shifts part of the trust burden rather than removing it. The correlated-false-corroboration case exposes that dependency directly.

This result complements, rather than competes with, existing CEAS work on rule-based FDIR [4] and AI-supported diagnosis and reconfiguration [5]. Those systems address how faults are represented, detected, diagnosed, or acted upon. Study 7 addresses a different assurance layer: whether the evidence feeding a recovery authorization decision is sufficient and trustworthy when an adversary can control an evidence producer. For learned aerospace recovery logic, validation should therefore trace not only model fit but also the provenance, observability, and failure relationships of decision-relevant evidence. This is consistent with aerospace machine-learning assurance guidance that treats confidence as a broader lifecycle evidence problem [3].

The limitations are substantial and deliberate. The learner is a deterministic integer linear-threshold model rather than a neural or stochastic learner. Inputs are binary, and the study does not model continuous or noisy evidence, stochastic learning, distribution shift, detector performance, or online adaptation. It supports no operational spacecraft, radio-frequency, flight, hardware, processor, energy, or latency claims. The corroboration input is designed to represent an independent evidence path in the prespecified model; operational independence is an assumption, not an empirical finding. The correlated-failure case demonstrates why that assumption matters.

These limitations constrain external validity but do not alter the finite-population claim. Every eight-feature visible state and every nine-feature corroboration state in the frozen design is evaluated. Future prospectively designed work could test the same information-sufficiency question with higher-fidelity evidence producers, noisy observables, learned representations, and empirically grounded mission trust paths.

6 Conclusion

The study shows that exact learning of a satellite cyber-recovery decision boundary does not establish correctness when the decisive truth is absent from the learner's observables. Adding corroboration can make a hidden-false state distinguishable, but the benefit disappears when the corroborating path shares the false state and can impose its own safety-versus-availability trade-off when evidence sources disagree.

The result is exact for the 1,033-position frozen model and does not establish operational spacecraft performance or certification sufficiency. Its assurance implication is narrower: validation of a learned recovery rule should include the information and trust architecture feeding that rule, not only the rule's fit to visible states.

Statements and Declarations

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Competing interests

The author has no relevant financial or non-financial interests to disclose.

Author contributions

Aman Kumar Singh is solely responsible for the conception and design of the study, implementation, experimental execution, analysis and verification, interpretation of results, manuscript preparation, and final approval of the submitted work.

Ethics approval

Not applicable. This study did not involve human participants, human data, animals, or biological material.

Consent to participate and consent for publication

Not applicable.

Data availability

The Study-7 protocol, canonical result records, policy summary, and separately implemented audit materials are publicly available in the mission-aware-satellite-cyber-recovery repository at <https://github.com/Zartharas/mission-aware-satellite-cyber-recovery>. The exact accepted Study-7 evidence is also preserved in a permanent Zenodo dataset deposit [9] (version DOI <https://doi.org/10.5281/zenodo.22732060>; concept DOI <https://doi.org/10.5281/zenodo.22732059>). The deposit preserves the accepted GitHub Actions artifact byte-for-byte and includes the exact observations, policy summary, trained-model parameters, execution report and provenance, implementation-independent repository audit output, and frozen protocol/results metadata. The accepted execution is identified by workflow run 33689625480 and commit f1530b0b2e81a5916adaf7ce808075156424dfb5; frozen SHA-256 identities are recorded in study7/results/RESULTS_FREEZE.json. No confidential, proprietary, or human-subject data are used.

Code availability

The protocol, implementation, analysis code, and separately implemented audit code are available at <https://github.com/Zartharas/mission-aware-satellite-cyber-recovery>. The scientific implementation paths and accepted execution identity are recorded in the repository freeze files.

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